

ECE253/CSE208 Introduction to Information Theory

Lecture 12: Channel Coding & Separation Theorem

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- Chap 7 of *Elements of Information Theory (2nd Edition)* by Thomas Cover & Joy Thomas
- Vembu-Verdu-Steinberg, The Source-Channel Separation Theorem Revisited, *IEEE Trans. on Information Theory*, Jan. 1995.

Repetition Codes

Repetition Codes

Type of Linear Codes

Hamming Codes

Joint

Source-Channel Coding Theorem

Source-Channel Separation Theorem

Source/Channel Coding Rates

Errors in communications can be detected and/or corrected by using

1. **Automatic Repeat Request (ARQ):** When a receiver circuit detects errors in a block of data, it requests that the block be re-transmitted.
2. **Forward Error Correction (FEC):** The transmitted data are encoded so that the receiver can correct as well as detect errors.

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Repetition Codes: Introduce redundancy naively by repeating the codes a few times.

- For example, consider $\{1 \rightarrow 11111; 0 \rightarrow 00000\}$. Do majority vote for decoding so that errors can be corrected as long as no more than 3 bits are corrupted. But this is not an efficient scheme since the rate is only $\frac{1}{5}$ bit per symbol.
- Repetition code cannot guarantee error-free communication, unless the effective rate converges to zero asymptotically.

Issues of Random Coding

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- The channel coding theorem promises the existence of good codes for transmitting information reliably at rates below capacity.

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- The channel coding theorem promises the existence of good codes for transmitting information reliably at rates below capacity.
- The direct part implies that when the block length n is long enough, if the codewords are chosen randomly, most likely the code is good.

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- A randomly constructed code has the following issues:

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 1. Encoding and decoding are computationally prohibitive (exponential in n).

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 2. High storage requirements for encoder and decoder.

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- A randomly constructed code has the following issues:
 1. Encoding and decoding are computationally prohibitive (exponential in n).
 2. High storage requirements for encoder and decoder.
 3. For example, if $R = \frac{1}{2}$ and $n = 1000$, we have $M = 2^{nR} = 2^{500} \approx 10^{150}$ codewords!

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- The construction of codes with efficient encoding and decoding algorithms is studied in *channel coding theory* (typically over finite fields).

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- The goal is to design capacity-achieving codes with computationally and storage-efficient encoding and decoding.
- Most practical channel codes are linear. A linear code is an algebraic code over a finite field such that the symbol-wise sum of two codewords and the scalar multiplication of a codeword by a field element are also codewords.

Type of Linear Codes: Block Codes

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Linear codes include block codes and convolutional codes.

- **Block codes:** Map a block of info bits into a codeword, no dependence on past bits. A binary block code with minimum distance d can correct $\lfloor \frac{d-1}{2} \rfloor$ bit errors.

Examples:

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Examples:

1. Hamming codes, cyclic codes, Reed-Solomon (RS) codes, Bose-Chaudhuri-Hocquenghem (BCH) codes, etc.

Applications: Satellite communications, storage devices such as DVD, SSD, USB, as well as high-speed applications like DVB, ADSL, xDSL.

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Applications: Satellite communications, storage devices such as DVD, SSD, USB, as well as high-speed applications like DVB, ADSL, xDSL.
2. Low-density parity-check (LDPC) codes and Polar codes.
Applications: 5G, DVB-S2, WiMAX, WiFi 4.

Type of Channel Codes: Convolutional Codes

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- **Convolutional codes:** Channel codes with *memory*, where each output block depends on the current input and several past inputs.

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- The encoder can be represented by a *finite-state machine* and a *trellis diagram*, which captures all possible state transitions over time.

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 - Turbo codes (3G/4G cellular systems, deep-space communications).

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 - Trellis-coded modulation (TCM) (digital modems, wireless and satellite communications).

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 - Turbo codes (3G/4G cellular systems, deep-space communications).
 - Trellis-coded modulation (TCM) (digital modems, wireless and satellite communications).
 - Trellis-based detection in magnetic and optical storage channels.

Development of Channel Codes

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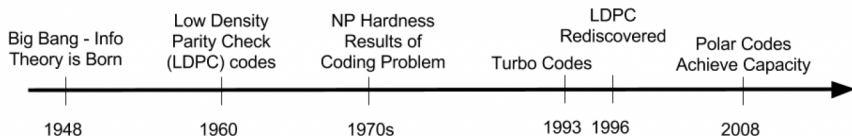


Figure: Timeline of developments towards efficient capacity-achieving codes. LDPC can reach within 0.0045 dB of the Shannon limit for binary AWGN channels. Figure credit to Prof. David Tse.

- Turbo codes were patented in April 1991 by Claude Berrou. The first public paper, “Near Shannon Limit Error-correcting Coding and Decoding: Turbo-code,” was presented at ICC 1993 by Berrou, Glavieux, and Thitimajshima.
- LDPC codes were first proposed by Robert Gallager in his 1962 PhD thesis at MIT.
- LDPC codes may be more efficient on relatively higher code rates (e.g. $3/4$, $5/6$, $7/8$) while turbo codes are the best solution at the lower code rates (e.g. $1/6$, $1/3$, $1/2$).

Hamming Codes: Background¹

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To illustrate key ideas underlying linear codes, we focus on Hamming codes.

Background:

- In the late 1940s, Richard Hamming worked on the Bell Model V relay computer using punched paper tape for input.

¹https://en.wikipedia.org/wiki/Hamming_code

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- Frustrated by having to restart programs after detected errors, he asked: “If a machine can detect an error, why can’t it locate and correct it?”

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- Frustrated by having to restart programs after detected errors, he asked: “If a machine can detect an error, why can’t it locate and correct it?”
- This led to his development of error-correcting codes. In 1950 he published the *Hamming code*, still used today in applications such as error-correcting code (ECC) memory.

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Hamming Codes: The Publication

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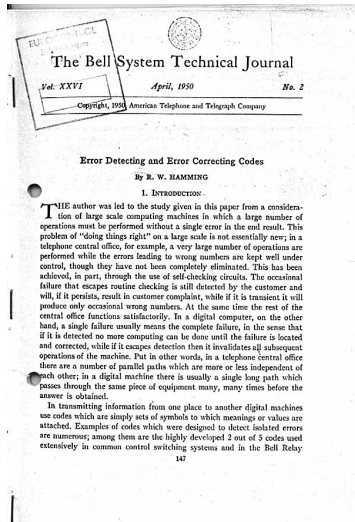
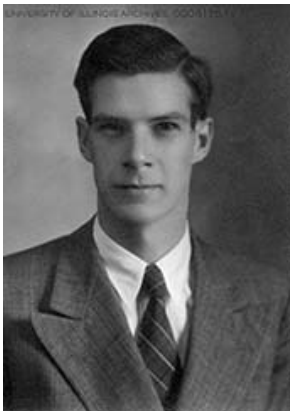
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(n, k, d) Hamming Code

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Parameters: $(n, k, d) = (7, 4, 3)$

- Encode $k = 4$ information bits into $n = 7$ bits.
- Minimum distance $d = 3$: minimum number of positions in which two codes differ.
- Minimum weight is 3: minimum number of 1's in any code (except for the all-zero).
- Can correct *one-bit errors*.

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Generalization: For $m \geq 2$ parity bits, the (n, k, d) Hamming code has parameters:

- block length $n = 2^m - 1$
- message length $k = 2^m - m - 1$
- minimum distance is always $d = 3$
- code rate is $R = \frac{k}{n} = 1 - \frac{m}{2^m - 1}$

Hamming Parity-check Matrix $\mathbf{H}$

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Parity-check matrix:

$$\mathbf{H}_{3 \times 7} = [\mathbf{P}^T \mid \mathbf{I}_3] = \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

- Columns of $\mathbf{H}$ are all nonzero binary vectors of length 3.
- $\text{rank}(\mathbf{H}) = 3$, $\dim(\text{null}(\mathbf{H})) = 4$.
- In general, $\mathbf{H}_{m \times n}$ is constructed by listing all non-zero vectors of length m .

Nullity Condition for Codewords

A valid codeword $\mathbf{c} = [c_1, c_2, \dots, c_7]$ must satisfy the parity-check condition

$$\mathbf{H}\mathbf{c}^T = \mathbf{0} \pmod{2}$$

Thus, the codewords form the null space of the parity-check matrix.

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Thus, the codewords form the **null space of the parity-check matrix**.

The $\mathbf{H}$ on the previous page yields codeword $\mathbf{c} = [d_1, d_2, d_3, d_4, p_1, p_2, p_3]$, consisting of four information bits followed by three parity bits. Hence, rows of $\mathbf{H}$ define parity constraints:

$$d_1 + d_2 + d_4 + p_1 = 0 \pmod{2} \Rightarrow p_1 = d_1 \oplus d_2 \oplus d_4,$$

$$d_1 + d_3 + d_4 + p_2 = 0 \pmod{2} \Rightarrow p_2 = d_1 \oplus d_3 \oplus d_4,$$

$$d_2 + d_3 + d_4 + p_3 = 0 \pmod{2} \Rightarrow p_3 = d_2 \oplus d_3 \oplus d_4.$$

Interpretation: each equation enforces even parity on a subset of bits.

Hamming Generator Matrix $\mathbf{G}$

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Generator matrix:

$$\mathbf{G}_{4 \times 7} = [\mathbf{I}_4 \mid \mathbf{P}] = \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right]$$

satisfies $\mathbf{H}\mathbf{G}^T = \mathbf{0}$. That is, $\text{rowspace}(\mathbf{G}) = \text{null}(\mathbf{H})$.

In principle we can choose any basis of the null space of $\mathbf{H}$ to construct $\mathbf{G}$. The above **systematic form** is convenient because

- the information bits appear explicitly in the codeword
- encoding becomes simple
- decoding explanations are clearer

Hamming Encoding

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Encoding rule

$$\mathbf{c} = \mathbf{u}\mathbf{G}$$

- $\mathbf{u} \in \mathbb{B}^4$: information bits; $\mathbf{c} \in \mathbb{B}^7$: transmitted codeword

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Example

$$\mathbf{c} = \mathbf{uG} = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \underbrace{1 & 0 & 1 & 1}_{\text{information bits}} & \underbrace{0 & 1 & 0}_{\text{parity bits}} \end{bmatrix}$$

(7,4,3) Hamming Code

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Since there are $2^k = 16$ possible $\mathbf{u} \in \mathbb{B}^k$, 16 codewords can be generated, as shown below.

0000000	0100101	1000011	1100110
0001111	0101010	1001100	1101001
0010110	0110011	1010101	1110000
0011001	0111100	1011010	1111111

Figure: All (7,4,3) Hamming codes.

Syndrome Decoding

Suppose the received vector is corrupted only in one bit (e.g., in bit i)

$$\mathbf{r} = \mathbf{c} + \mathbf{e}_i,$$

where $\mathbf{e}_i$ is the i -th canonical vector.

Compute the **syndrome**

$$\mathbf{s} = \mathbf{H}\mathbf{r}$$

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- If $\mathbf{s} = \mathbf{0} \rightarrow$ no error.
- *Error correction:*

$$\mathbf{s} = \mathbf{H}(\mathbf{c} + \mathbf{e}_i) = \mathbf{H}\mathbf{e}_i = \mathbf{H}(:, i) \leftarrow \text{the } i\text{-th column of } \mathbf{H},$$

Then, the error is in bit i .

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Thus, the Hamming code can **locate and correct one-bit errors**.

(7,4,3) Hamming Code: Venn Diagram

Repetition
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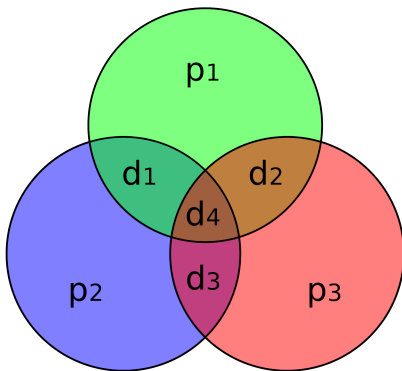
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- The four information bits d_1, d_2, d_3, d_4 occupy the intersection regions, while the three parity bits p_1, p_2, p_3 correspond to the three circles.
- Each circle represents one parity-check constraint requiring even parity.

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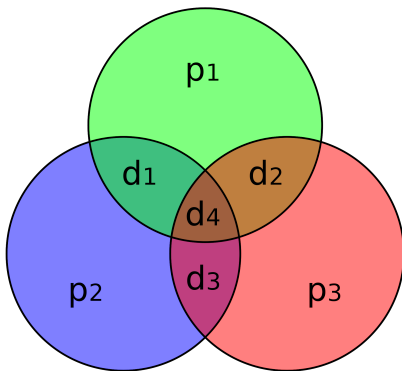
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- Each circle represents one parity-check constraint requiring even parity.
- The diagram assumes the classical bit ordering
$$(c_1, c_2, c_3, c_4, c_5, c_6, c_7) = (p_1, p_2, d_1, p_3, d_2, d_3, d_4).$$
Parity bits are placed at positions 1, 2, 4 because these correspond to powers of two.
- Different bit orderings yield equivalent parity checks but may permute the columns of $\mathbf{H}$ and $\mathbf{G}$ accordingly.

All Possible Hamming Codes

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Parity bits	Total bits	Data bits	Name	Rate
2	3	1	Hamming(3,1) (Triple repetition code)	$1/3 \approx 0.333$
3	7	4	Hamming(7,4)	$4/7 \approx 0.571$
4	15	11	Hamming(15,11)	$11/15 \approx 0.733$
5	31	26	Hamming(31,26)	$26/31 \approx 0.839$
6	63	57	Hamming(63,57)	$57/63 \approx 0.905$
7	127	120	Hamming(127,120)	$120/127 \approx 0.945$
8	255	247	Hamming(255,247)	$247/255 \approx 0.969$
...				
m	$n = 2^m - 1$	$k = 2^m - m - 1$	Hamming($2^m - 1, 2^m - m - 1$)	$(2^m - m - 1)/(2^m - 1)$

Setup: Entropy Rate and Channel Capacity

Setup

- Source: stationary ergodic finite-alphabet process

$$V_1, V_2, \dots, V_n$$

with entropy rate

$$H(\mathcal{V}) = \lim_{n \rightarrow \infty} \frac{1}{n} H(V^n)$$

- Channel: DMC with capacity

$$C = \max_{p(x)} I(X; Y)$$

Under this setup, the fundamental condition for reliable communication is

$$H(V) < C$$

Joint Source–Channel Coding (JSCC) Theorem

Repetition
Codes

Type of Linear
Codes

Hamming Codes

Joint
Source–Channel
Coding Theorem

Source–Channel
Separation
Theorem

Source/Channel
Coding Rates

Theorem

There exists a sequence of source-channel encoders and decoders

$$V^n \rightarrow X^n \rightarrow Y^n \rightarrow \hat{V}^n$$

such that

$$\Pr(\hat{V}^n \neq V^n) \rightarrow 0$$

if and only if

$$H(\mathcal{V}) < C.$$

If $H(\mathcal{V}) > C$, the probability of error is bounded away from zero.

Shannon Source–Channel Separation Theorem

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Theorem

If $H(\mathcal{V}) < C$, then reliable communication is achievable by

- *compressing the source to any rate $R > H(\mathcal{V})$*
- *transmitting using a channel code with $R < C$*

Choose

$$H(\mathcal{V}) < R < C$$

to obtain

$$\Pr(\hat{V}^n \neq V^n) \rightarrow 0.$$

Joint Coding vs Separation

Joint Source–Channel Coding

$$V^n \rightarrow X^n \rightarrow Y^n \rightarrow \hat{V}^n$$

- Encoder directly maps source sequence to channel inputs
- Allows fully joint design
- Fundamental reliability condition $H(\mathcal{V}) < C$

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Joint Coding vs Separation

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Joint Source-Channel Coding

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Source-Channel Separation

$$V^n \rightarrow \text{Source Encoder} \rightarrow \text{Channel Encoder} \rightarrow X^n$$

- Two independent coding stages
- Simpler design and analysis
- Achieves the same optimal limit $H(\mathcal{V}) < C$

When Is Source–Channel Separation Optimal?

Engineering implications: Two-stage is as good as one-stage process.

- Asymptotic optimality can be achieved by separating source and channel coding.
- Design source codes for the most efficient representation of the data.
- *Separately and independently* design channel codes appropriate for the channel.

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Engineering implications: Two-stage is as good as one-stage process.

- Asymptotic optimality can be achieved by separating source and channel coding.
- Design source codes for the most efficient representation of the data.
- *Separately and independently* design channel codes appropriate for the channel.

Source–channel separation is asymptotically optimal for

- stationary ergodic finite-alphabet sources
- discrete memoryless channels (DMC)
- point-to-point communication
- arbitrarily large blocklength (no delay constraint)

The optimality of source–channel separation is not universal. Outside the aforementioned regime, separate source and channel coding may no longer be optimal.

When Is Joint Coding Better?

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Source/Channel
Coding Rates

Joint source–channel coding can provide better performance in

- channels with memory or state (e.g., fading channels)
- non-ergodic sources or channels
- multiuser networks
- finite blocklength or delay-constrained systems

When Is Joint Coding Better?

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Example.

- Sending English text over an erasure channel: corrupted bits are difficult to decode.
- Redundancy in the source is suited to the channel: speech for the human ear.

Source Coding Rate R_s

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Coding Rates

Suppose a block of m source symbols is compressed into k bits.

Source coding rate

$$R_s = \frac{k}{m} \quad (\text{bits per source symbol})$$

Source coding theorem

For lossless compression:

$$R_s \geq H(X)$$

Interpretation

R_s represents the average number of bits required to represent each source symbol.

Average Code Length $\bar{L}$ and Source Coding Rate R_s

Let $s \in \mathcal{S}$ denote a source symbol from the source alphabet $\mathcal{S}$. For a variable-length source code with codeword lengths $\ell(s)$, the **average code length** is given as

$$\bar{L} = \sum_{s \in \mathcal{S}} p(s) \ell(s) \quad (\text{bits per source symbol})$$

- $\bar{L}$ measures the expected number of bits used to encode one source symbol.

Relation to source coding rate

- If symbols are encoded independently using a variable-length code,

$$R_s = \bar{L}.$$

- If blocks of length m are encoded with average code length $\bar{L}_m$,

$$R_s = \frac{\bar{L}_m}{m}.$$

Channel Code Block Length n

In channel coding, we encode k information bits into a codeword of length n :

$$x_1^n := (x_1, \dots, x_n) \in \mathcal{X}^n$$

Important clarification

n = number of channel uses.

Each channel use corresponds to sending one symbol x_i from the channel input alphabet $\mathcal{X}$.

Examples:

- Binary channel: one bit per channel use
- q -ary channel: one symbol from $\{1, \dots, q\}$ per channel use
- AWGN channel: one real-valued symbol per channel use

Thus the block length n equals the number of times the channel is used.

Channel Coding Rate R_c

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Hamming Codes

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Coding Theorem

Source-Channel
Separation
Theorem

Source/Channel
Coding Rates

Suppose k information bits are encoded into a codeword of length n .

Channel coding rate

$$R_c = \frac{k}{n} \quad (\text{bits per channel use})$$

Shannon channel coding theorem

Reliable communication is possible only if

$$R_c < C$$

where C is the channel capacity.

Bandwidth Expansion Factor B and Source–Channel Rate ρ

Repetition
Codes

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Source/Channel
Coding Rates

Suppose m source symbols are sent via n channel uses.

Bandwidth expansion factor

$$B = \frac{n}{m} \quad (\text{channel uses per source symbol})$$

Source–channel rate

$$\rho = \frac{1}{B} = \frac{m}{n} \quad (\text{source symbols per channel use})$$

These quantities describe how source symbols are mapped to channel uses.

Feasibility Condition for Reliable Communication

Repetition
Codes

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Coding Rates

Combining the source coding theorem and channel coding theorem:

Source compression:

$$R_s \geq H(X)$$

Channel reliability:

$$R_c \leq C$$

Using

$$R_s = \frac{k}{m}, \quad R_c = \frac{k}{n}, \quad B = \frac{n}{m} = \frac{R_s}{R_c}$$

we obtain

$$H(X) \leq BC$$

This is the fundamental condition for reliable communication.

Interpretation

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$$H(X) \leq BC$$

- $H(X)$: information produced by the source per symbol
- C : maximum information the channel can transmit per channel use
- B : number of channel uses per source symbol

Meaning

source information rate $\leq$ channel transmission capability

Reliable communication is possible if the channel can carry at least as much information as the source generates.